

数学作业纸

科目_____

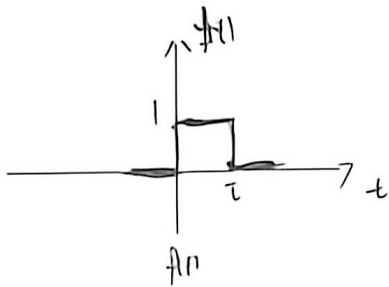
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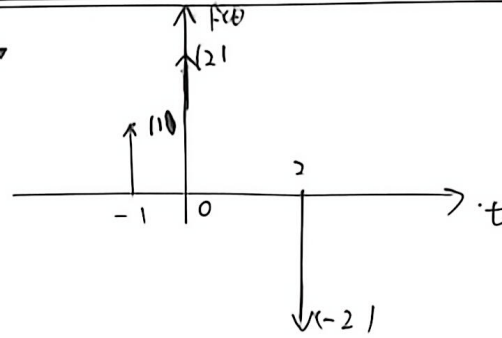
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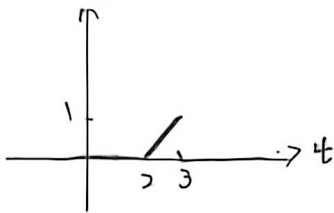
1-3 (2)



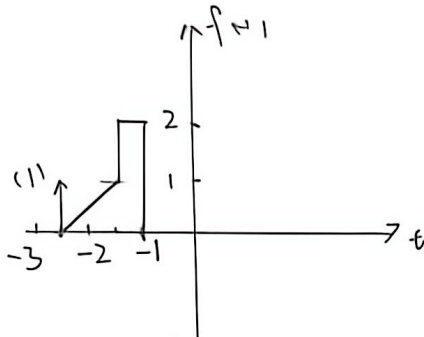
(5)



1-5 (3)



1-7 (2)



1-8 (2). $f(0+2) = 1$ (4) $2 + e^{-4}$ (b) $\int_{-1}^3 e^{j\omega t} \delta(t) dt = \int_{-1}^3 e^{-j\omega t} \int_{t-2}^t \delta(t-2) dt$
 $e^0 = 1$
 $\cdot \frac{1}{t} = e^{-j\omega \cdot 2}, e^{-j\omega \cdot 2}$

1-11 (2). $r_1(t) = 4 \ln[2e^{j\omega t}] u(t)$

$$e^{j\omega t} = a e^{j\omega t}$$

$$r_2(t) = 4 \ln[a e^{j\omega t}] u(t)$$

$$a r_1(t) = a 4 \ln[e^{j\omega t}] u(t). \text{ 非线性. }$$

5. 假设初始状态 $r(0) = 0$. $r_2(t) = 2t e^{j\omega t}$.

假设输入 $e^{j\omega t}$. $r_2(t) = r_2(0)$.

设初始状态为 $x_1 = r(0)$. 重新输入 x_1^2 .

$$r^2 x_1^2 \neq r(x_1^2). \text{ 非线性. }$$

$$r^2 x_1^2 \neq r(x_1^2). \text{ 非线性. }$$

1-12

$$e_1(t) = e^{j\omega(t-t_0)} \quad r_1(t) = e^{j\omega t} u(t) = e^{j\omega(t-t_0)} u(t).$$

$$r(t-t_0) = e^{j\omega(t-t_0)} u(t-t_0). \text{ 非时不变. }$$

$$e_1(t) = e^{j\omega(t-t_0)} \quad r_1(t) = 4 \ln(t) e^{j\omega(t-t_0)}$$

$$r_1(t) = 4 \ln(t-t_0) e^{j\omega(t-t_0)} \text{ 非时不变}$$

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5) $e_1(t) = e^{-(t-t_0)}$ $r_1(t) = e^{-(t-t_0)} + \frac{d}{dt} e^{-(t-t_0)} = e^{-(t-t_0)} + e^{-(t-t_0)}(-1) = r^{-(t-t_0)}$.
 $r_1(t) = r^{-(t-t_0)}$ 时变.

1-13 (1) $y_1(t) = e^{-(1-t)}$

$y_2(t) = e^{-(1-t)}$
 $y_3(t) = [ae^{-(1-t)} + be^{-(1-t)}] (1-t) = ae^{-(1-t)} + be^{-(1-t)}$.
 $y_3(t) = ay_1(t) + by_2(t)$. 时变.

1 $e^{-(t-t_0)} = e^{-(1-t-t_0)} = e^{-(1-t-t_0)}$
 $r^{-(t-t_0)} = e^{-(1-t-t_0)} = e^{-(1-t-t_0)}$.

时变.
 $r(0) = e^{-(1-0)} = e^{-1}$. $t=0$ 时输入. $t=1$ 时输入.
 非因果.

2) $y_3(t) = \int_{-\infty}^{5t} [ae^{-(\tau)} + be^{-(\tau)}] d\tau = a \int_{-\infty}^{5t} e^{-(\tau)} d\tau + b \int_{-\infty}^{5t} e^{-(\tau)} d\tau = a[1 - e^{-5t}] + b[1 - e^{-5t}]$
 时变.

$e^{-(t-t_0)} = e^{-(t-t_0)}$ $r^{-(t-t_0)} = \int_{-\infty}^{5t} e^{-(\tau)} d\tau = \int_{-\infty}^{5t} e^{-(\tau-t_0)} d\tau$. $\lambda = \tau - t_0$, $d\tau = d\lambda$.

$r^{-(t)} = \int_{-\infty}^{5t-t_0} e^{-(\lambda)} d\lambda$.

$r^{-(t-t_0)} = \int_{-\infty}^{5t-t_0} e^{-(\tau)} d\tau = \int_{-\infty}^{5t-5t_0} e^{-(\tau)} d\tau$. 时变.

$r(1) = \int_{-\infty}^5 e^{-(\tau)} d\tau$. $t=1$ 时输入. $t=5$ 时输入.
 非因果.

2-5 (4) $\frac{1}{s} \cdot \frac{1}{s^2 + 4s + 8} = \frac{1}{s} \cdot \frac{1}{(s+2)^2 + 4}$.

$\frac{1}{s} \cdot \frac{1}{(s+2)^2 + 4} = \frac{1}{s} \cdot \frac{1}{(s+2)^2 + 2^2} = \frac{1}{s} \cdot \frac{1}{(s+2)^2 + 2^2}$.

$h(t) = [4e^{-2t} \cos(2t) - 3e^{-2t} \sin(2t)] u(t)$.

$G(s) = \frac{H(s)}{s} = \frac{4s+2}{s(s^2+4s+8)}$.

$= \frac{A}{s} + \frac{B4+C}{s^2+4s+8}$.

$A(s^2+4s+8) + s(B4+C) = (s+1)$ $s^2 + (4A+B)s + (8A+4B+C) = (s+1)$ $4A+B=1$ $8A+4B+C=0$.

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$$\begin{cases} A+B=0 \\ 4A+C=4 \\ 8A=2 \end{cases} \Rightarrow \begin{cases} A=\frac{1}{4} \\ B=-\frac{1}{4} \\ C=3 \end{cases} \quad G(s) = \frac{1}{s} + \frac{-\frac{1}{4}s+3}{s^2+4s+8} = \frac{1}{s} + \frac{s-12}{(s+2)^2+2^2}$$

$$= \frac{1}{4} \left[\frac{1}{s} - \frac{s+2}{(s+2)^2+2^2} + \frac{14}{(s+2)^2+2^2} \right]$$

$$g(t) = \frac{1}{4} \left[1 - e^{-2t} (\cos(2t) + 7e^{-2t} \sin(2t)) \right] u(t)$$

$$2-6(1). \quad \Rightarrow R(s) = \frac{1}{s} - 2 + 3 \frac{s+1}{s^2+4s+8} = \frac{1}{s} + \frac{3s+5}{s^2+4s+8}$$

$$R(s) \left[\frac{s^2+3s+2}{s^2+4s+8} \right] = (s+5) + (4s+3) \frac{s+1}{s^2+4s+8}$$

$$R(s) = \frac{s+5}{s^2+4s+8} + \frac{4s+3}{s^2+4s+8} = \frac{5s+8}{s^2+4s+8}$$

$$1/(s) = \frac{s+3}{s^2+4s+8} = \frac{s+3}{(s+1)(s+2)} = \frac{K_1}{s+1} + \frac{K_2}{s+2}$$

$$K_1 = \frac{s+3}{s+2} \Big|_{s=-1} = 2$$

$$h(t) = (2e^{-t} - e^{-2t}) u(t)$$

$$K_2 = \frac{s+3}{s+1} \Big|_{s=-2} = -1$$

$$R(s) = \frac{s+5}{s^2+4s+8} = \frac{s+5}{(s+1)(s+2)} = \frac{A}{s+1} + \frac{B}{s+2}$$

$$A = \frac{s+5}{s+2} \Big|_{s=-1} = 4$$

$$B = \frac{s+5}{s+1} \Big|_{s=-2} = -3$$

$$r_2(t) = (4e^{-t} - 3e^{-2t}) u(t)$$

$$E(s) = \frac{1}{s} \Rightarrow R(s) = H(s)E(s) = \frac{s+1}{s(s+1)(s+2)} = \frac{C}{s} + \frac{D}{s+1} + \frac{E}{s+2}$$

$$C = \frac{s+1}{s(s+2)} \Big|_{s=0} = \frac{1}{2} = 0.5$$

$$D = \frac{s+1}{s(s+2)} \Big|_{s=-1} = -2$$

$$r_3(t) = (0.5 - 2e^{-t} + 0.5e^{-2t}) u(t)$$

$$E = \frac{s+1}{s(s+1)} \Big|_{s=-2} = 0.5$$

$$r(t) = h_2(t) + r_3(t) = (1.5e^{-t} - 2.5e^{-2t}) u(t)$$

自由响应 $\frac{1}{2}(2e^{-t} - 2.5e^{-2t}) u(t)$

强迫响应 $1.5 u(t)$

总响应 $(2e^{-t} - 2.5e^{-2t}) u(t)$

稳态值 $1.5 u(t)$

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$$2-7 \quad 11) \cdot y_1(t) = \int_{-\infty}^{+\infty} e^{-\tau} u(\tau) u(t-\tau) d\tau = \int_0^t e^{-\tau} d\tau = [-e^{-\tau}]_0^t = -e^{-t} - (-1) = 1 - e^{-t}$$

$$y_1(t) = (1 - e^{-t}) u(t).$$

$$14) \cdot y_1(t) = [\delta(t+1) - \delta(t-1)] * \cos \omega t$$

$$= \cos(\omega t) * \delta(t+1) - \cos(\omega t) * \delta(t-1)$$

$$y_1(t) = \cos[\omega(t+1)] - \cos[\omega(t-1)]$$

$$= -2 \sin(\omega t) \sin(\omega)$$

$$15) \cdot y_1(t) = [u(t) - u(t-1)] * [u(t-2) - u(t-3)]$$

$$= u(t) * u(t-2) - u(t) * u(t-3) - u(t-1) * u(t-2) + u(t-1) * u(t-3)$$

$$u(t) * u(t-2) = (t-2) u(t-2)$$

$$u(t) * u(t-3) = (t-3) u(t-3)$$

$$u(t-1) * u(t-2) = (t-1-2) u(t-1-2) = (t-3) u(t-3)$$

$$u(t-1) * u(t-3) = (t-4) u(t-4)$$

$$y_1(t) = (t-2) u(t-2) - 2(t-3) u(t-3) + (t-4) u(t-4)$$

$$2-16 \quad h(t) = g'(t) = -4e^{-2t} u(t) - \delta(t)$$

$$r_{2,1}(t) = e(t) * h(t) = e^{-3t} u(t) * [-4e^{-2t} u(t) - \delta(t)] = -4 \int_0^t e^{-3\tau} e^{-2(t-\tau)} d\tau - e^{-3t} u(t)$$

$$= (3e^{-3t} - 4e^{-2t}) u(t)$$